LIN Gateway User Guide
V0.1

1 REVISION HISTORY

Table 1 Revision History

Rev #	Date	Action	By
0.1	11/05/2021	First draft	JackPan

2 TABLE OF CONTENTS

1 REVISION HISTORY 2

2 TABLE OF CONTENTS 3

3 SYSTEM OVERVIEW 4

3.1 Hardware connection 4

4 PC SOFTWARE GUI DESCRIPTION 5

4.1 Log and Debug 7

4.2 Command Editor 8

4.2.1 Command Content Edit 9

4.2.2 Save/Load commands to/from File 9

4.3 LIN Demo 10

4.4 Bootloader 10

4.5 Auto Addressing 11

4.6 Programming 11

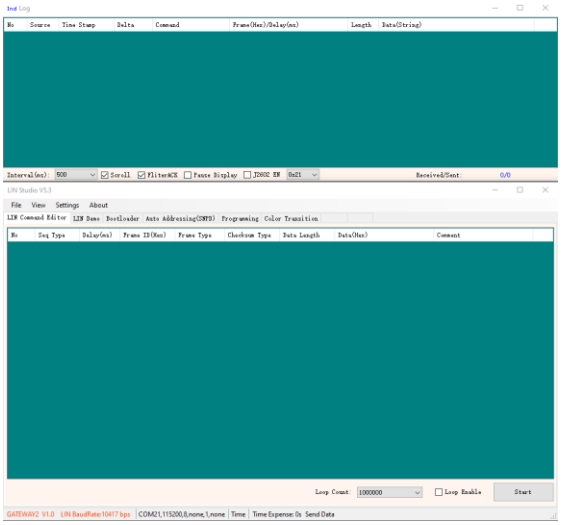
4.7 Color Transition 12

5 OFFLINE LIN COMMAND 13

6 COMMUNICATION PROTOCOL BETWEEN PC TOOL AND LIN GATEWAY 15

3 System Overview

LIN gateway is designed to speed up indiemicro interior light and LIN bus-based production development, it integrates general LIN command editor, interior light color control and color configuration, bootloader, chip programming, and general color transition formula etc.



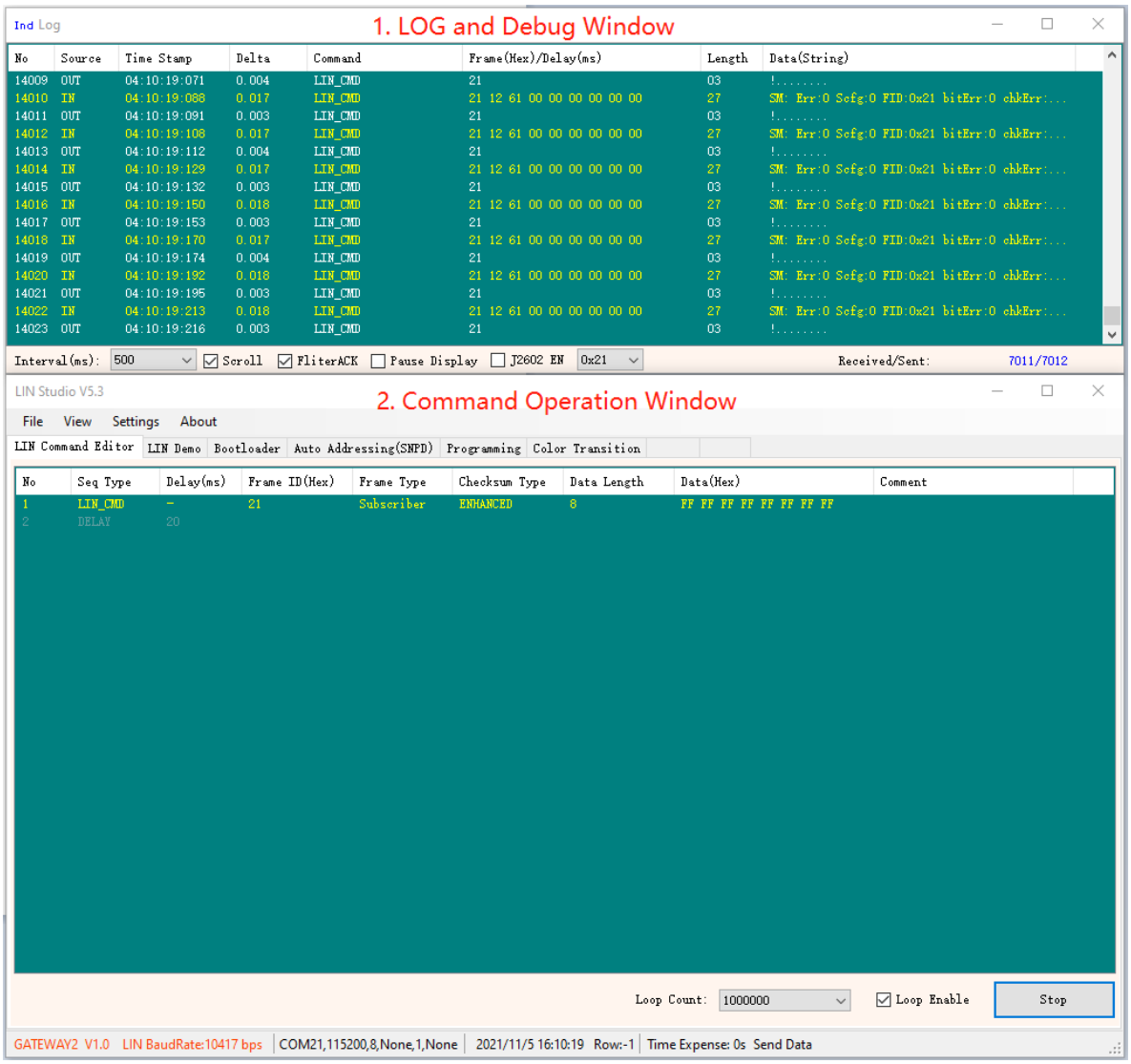
3.1 Hardware connection

LIN gateway is a LIN master, it transfers the data from PC software command to LIN slave and send back the LIN slave data to PC software.

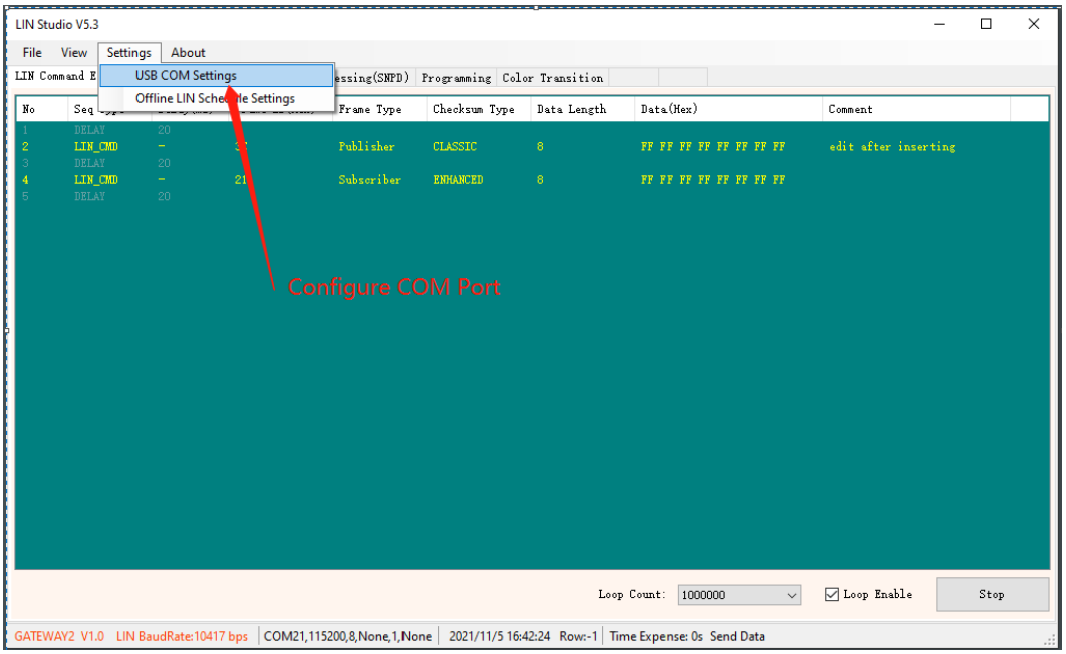
4 PC SOFTWARE GUI DESCRIPTION

1) ***LOG and Debug Window***: used to log all of the command and data which operated through Command Operation Window.

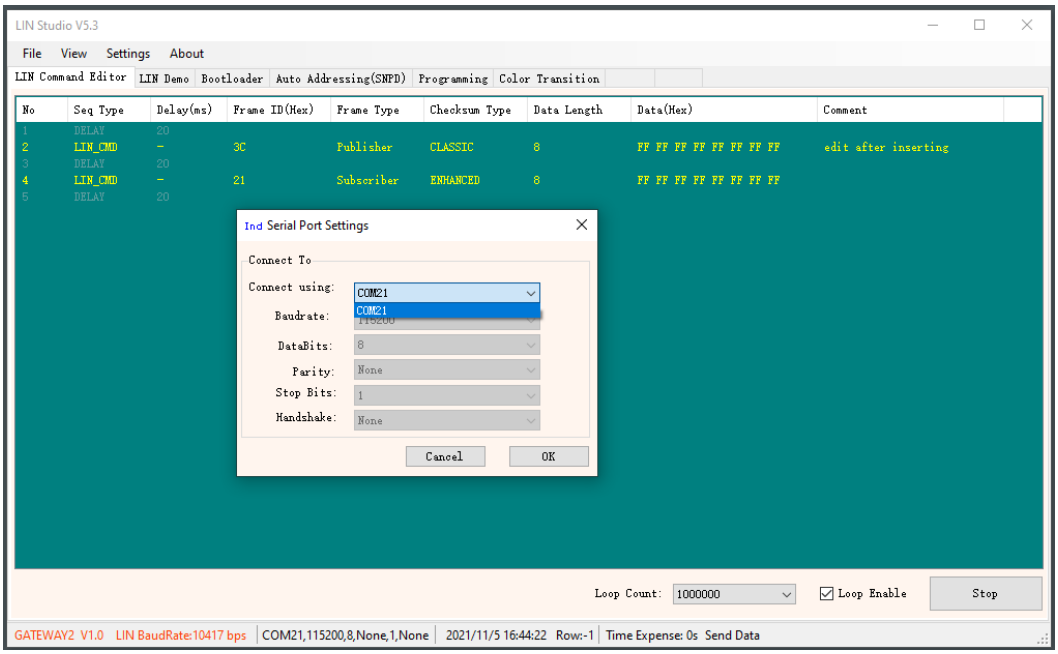
2) ***Command Operation Window***: all of the commands are defined in this window.



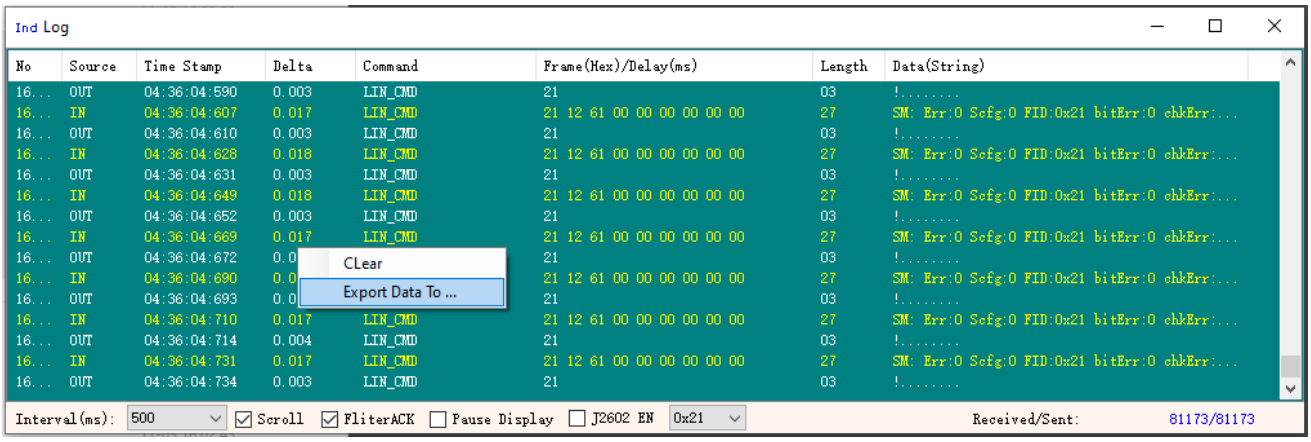
Communication Port need to be set before communicating with LIN gateway:



Select the gateway COM port and press OK button:



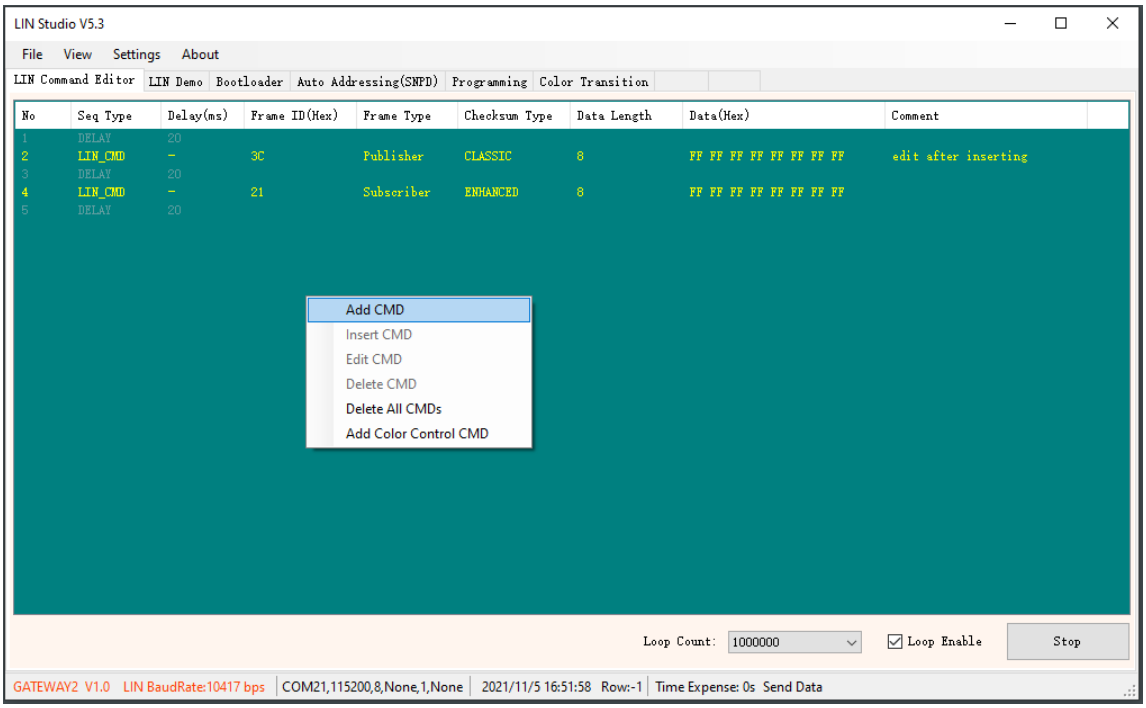
4.1 LOG AND DEBUG



1) ***Interval(ms)***: data display refresh rate.
2) ***Scroll***: always display the latest information when enabled.
3) ***FilterACK***: there is handshake command between PC software and LIN gateway, disabled to check whether the communication is okey or not.
4) ***Pause Display***: pause display to help browse the log data
5) ***J2602 EN***: enabled to interpret J2602 command, if disabled, the developer can pick up the LIN2.2 state report command FID through the dropdown list (FID= 0x21 by default)
6) ***Received/Sent***: Sent data and got data statistics.

Right click to save the data log or clear the log window.

4.2 COMMAND EDITOR

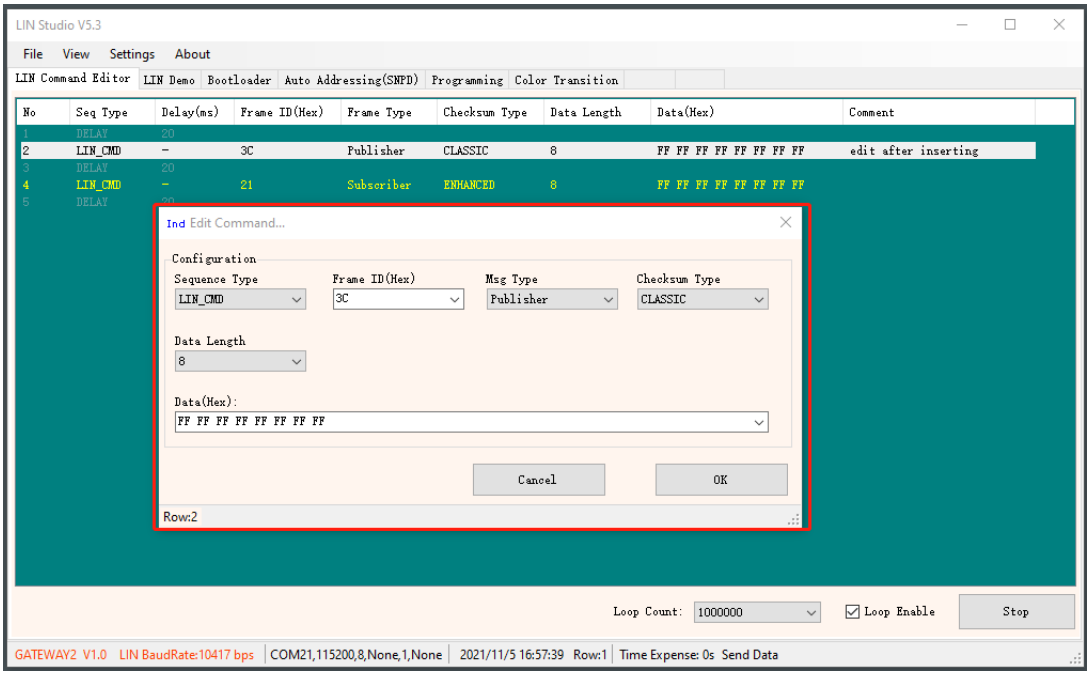


Edit LIN command by right clicking LIN command Editor page

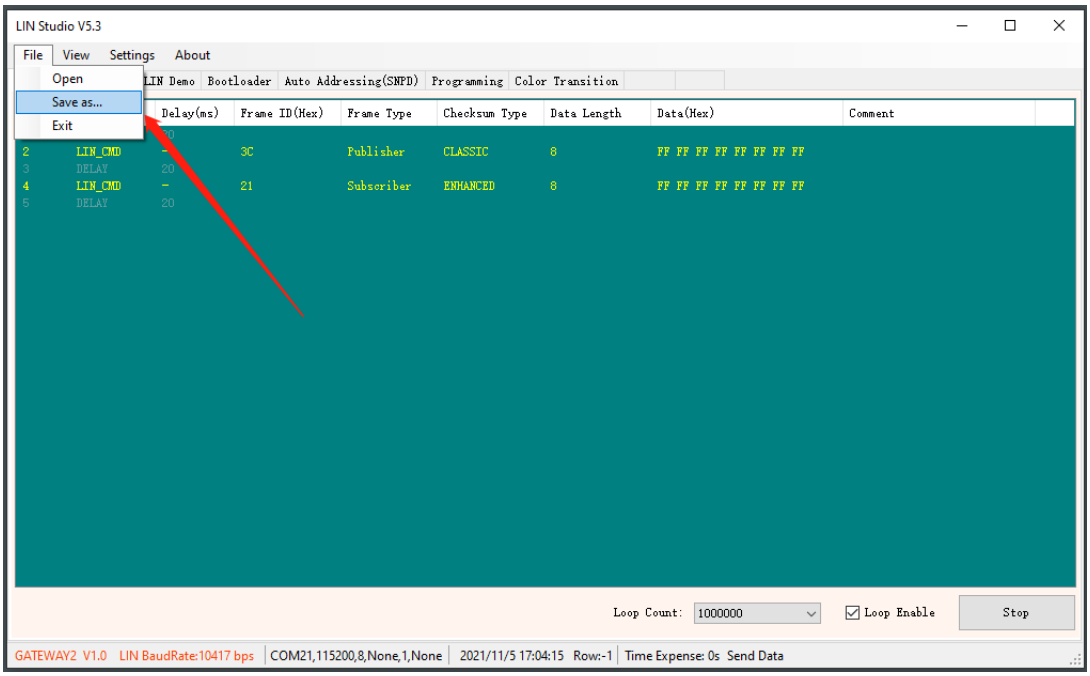
1) ***Add CMD*** : Add LIN command or Add a delay time between LIN commands
2) ***Insert CMD***: Insert command between exist commands.
3) ***Edit CMD***: Edit exist command
4) ***Delete all CMDs***: Clear all of the command in the command editor.
5) ***Add Color Control CMD***: Add the color control command which is defined in SDK.

4.2.1 Command Content Edit

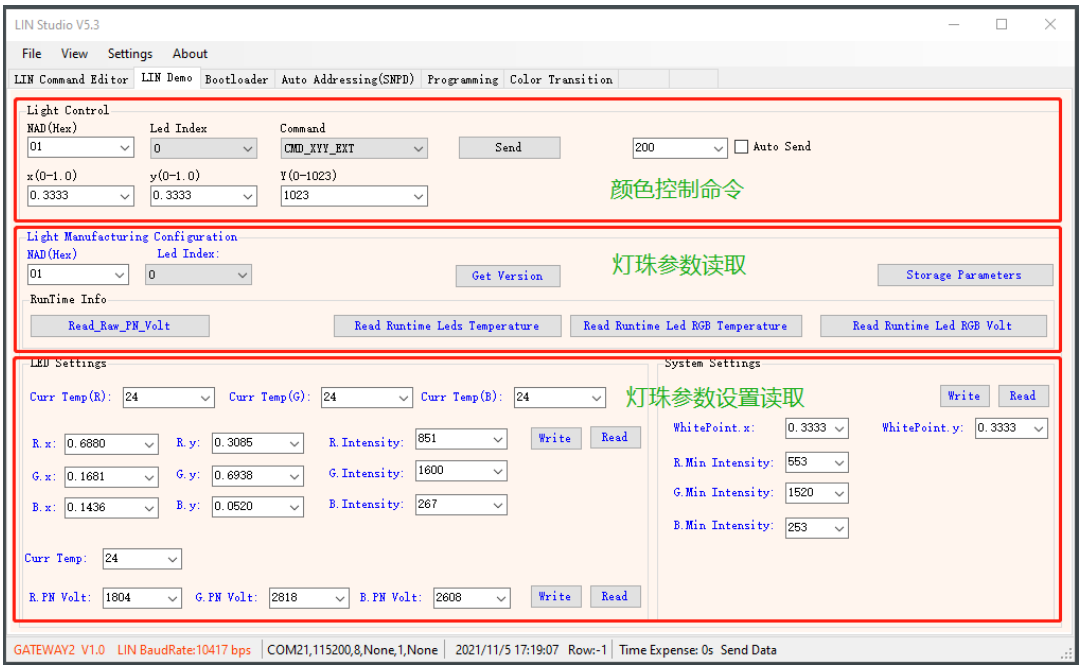
Click one of the commands to enter command configuration window:



4.2.2 Save/Load commands to/from File

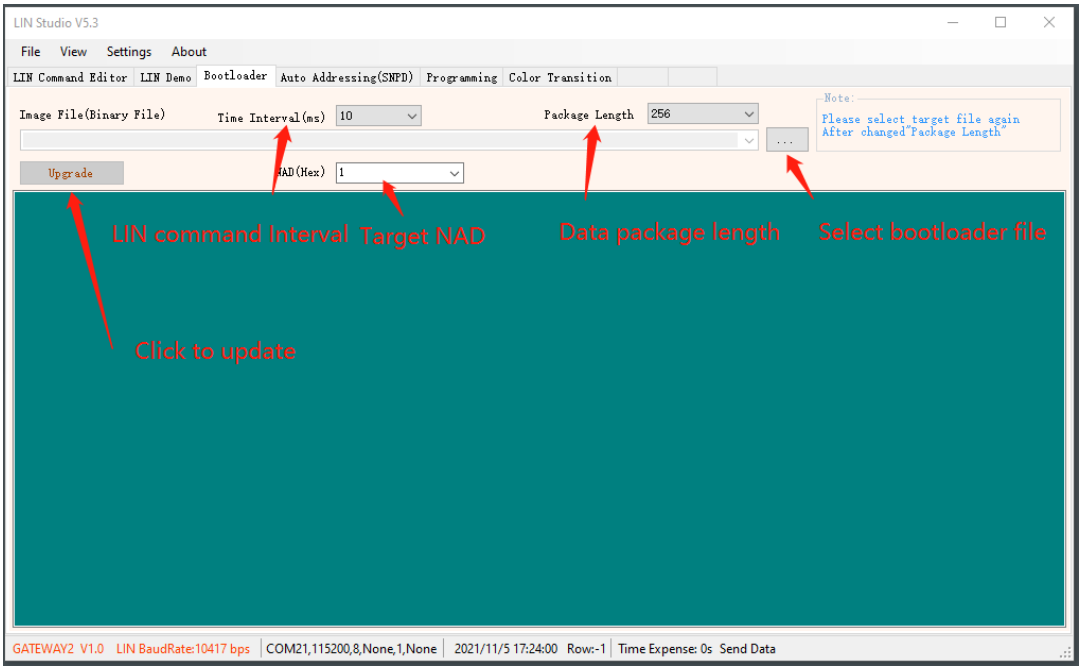


4.3 LIN DEMO

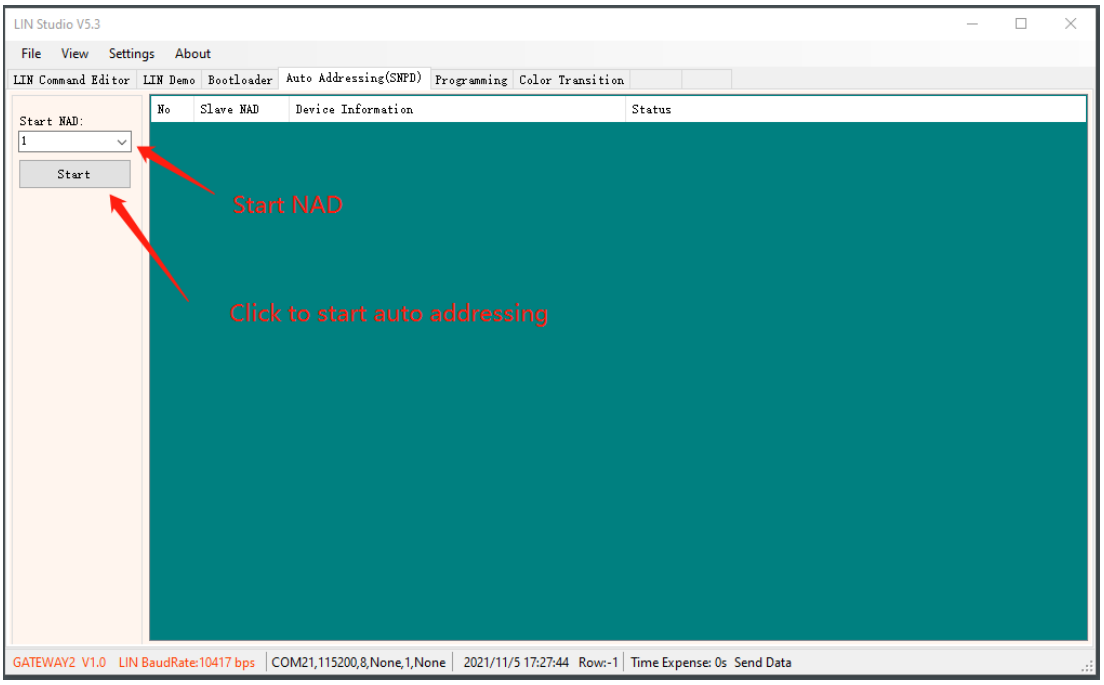


Refers to Color calibration user guide for specific information.

4.4 BOOTLOADER

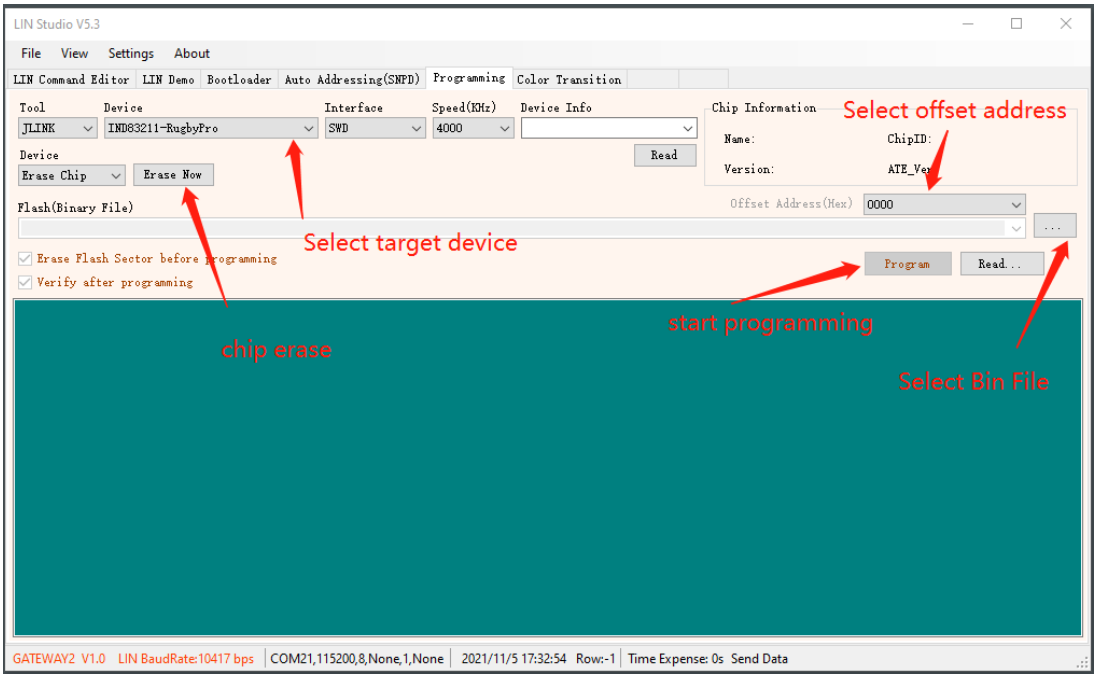


4.5 AUTO ADDRESSING

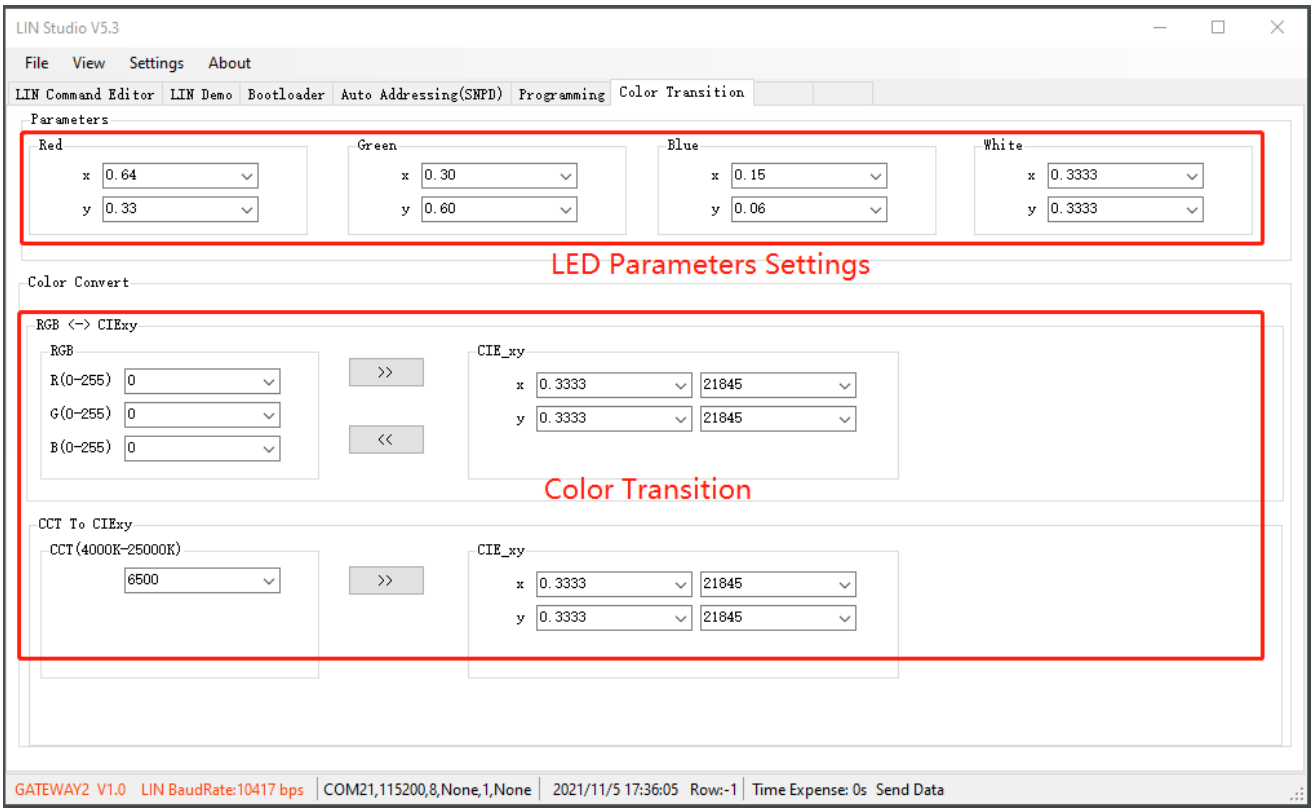


4.6 PROGRAMMING

Current PC tool supports programming the target device together with Jlink tool.

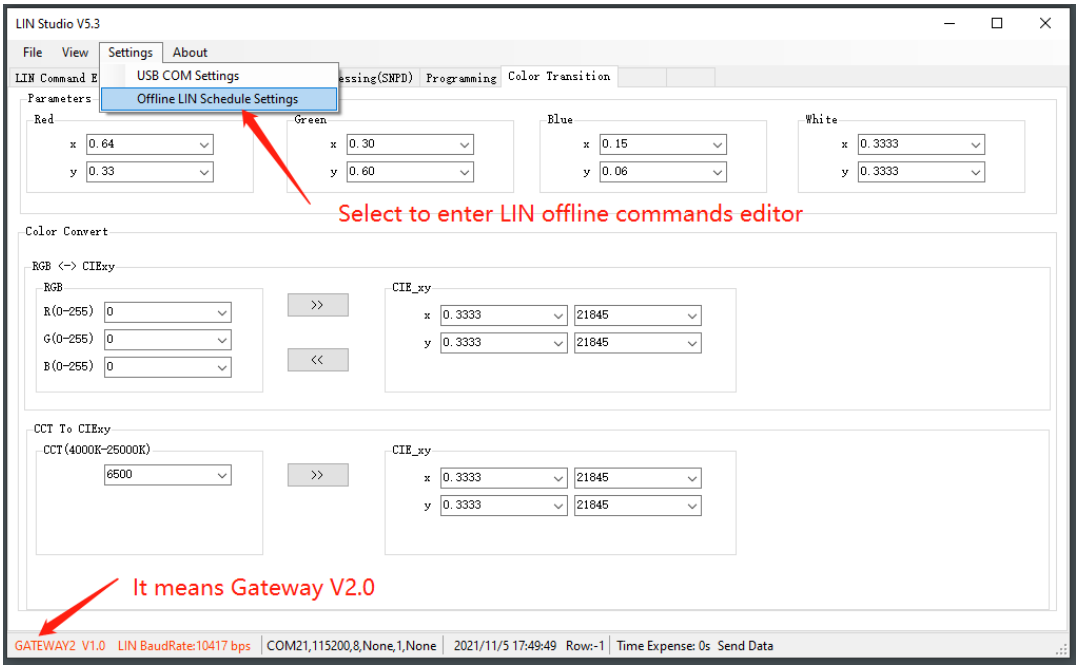


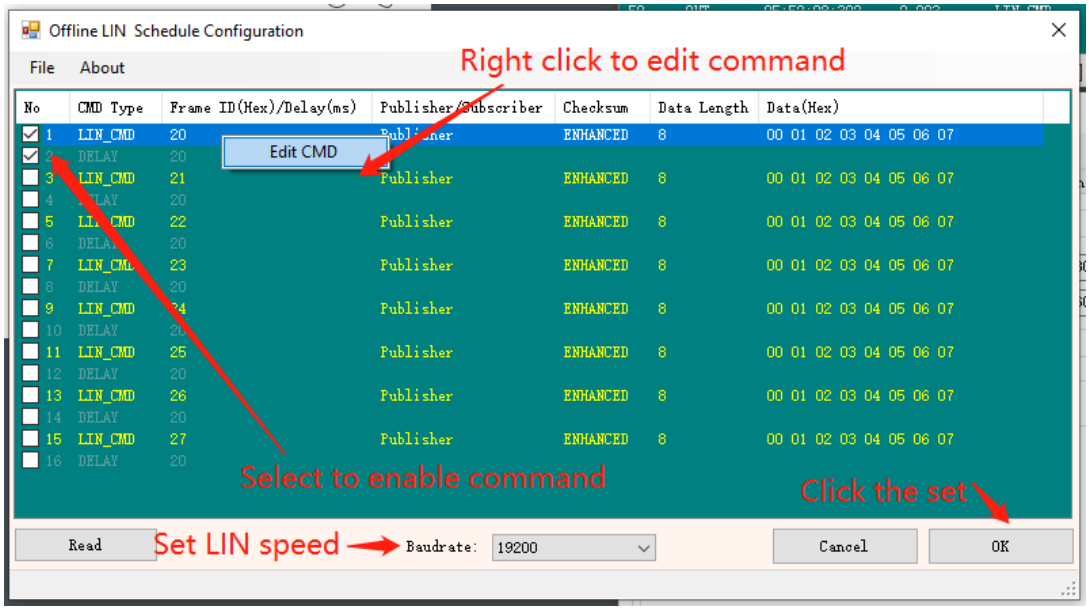
4.7 Color Transition



5 OFFLINE LIN COMMAND

There is an offline command editor in PC software which can run in gateway without PC tool, this function is only supported by LIN gateway V2.0.





Press “START” button to run offline command periodically, Press “STOP” button to stop offline command.

6 COMMUNICATION PROTOCOL BETWEEN PC TOOL AND LIN GATEWAY

COM Settings:

BaudRate:	115200
DataBits:	8
Parity	None
Stop Bits	1

Data Frame Description

Head(1byte)	FrameID(1byte)	PEC(1byte)	Control Byte(1byte)			Data(Data Lengthbyte)
			Data Length:bit7:4	Frame Type:bit3:1	Checksum Type: bit0	Data[0:Data Length-1]

Head:	0x55:	SEND	Frame ID	0x00-0x3F	LIN Frame ID without Parity
	0xA5:	ACK/RSP			
PEC	Sum of Control Byte + Data		Control Byte		
			Data Length	0-8	Data size
			Frame Type	0x00:	Request Data
				0x01:	Send data
			Checksum Type	0x00:	Classic
				0x01:	Enhanced

Send Data To Target

	Head(1byte)	FrameID(1byte)	PEC(1byte)	Control Byte(1byte)			Data(Data Lengthbyte)
				Data Length:bit7:4	Frame Type:bit3:1	Checksum Type: bit0	Data[0:Data Length-1]
SEND	**0x55**	**FID**		**Len**	**1**	**Checksum**	**Data[0:Len-1]**
	Head(1byte)	FrameID(1byte)	PEC(1byte)	Control Byte(1byte)			Data(Data Lengthbyte)
				Data Length:bit7:4	Frame Type:bit3:1	Checksum Type: bit0	
ACK	**0xA5**	**FID**		0	1	Checksum	

Request Data From Target

	Head(1byte)	FrameID(1byte)	PEC(1byte)	Control Byte(1byte)			Data(Data Lengthbyte)
				Data Length:bit7:4	Frame Type:bit3:1	Checksum Type: bit0	Data[0:Data Length-1]
Request	**0x55**	**FID**		**Len**	**0**	**Checksum**	**Data[0:Len-1]=0xFF**
	Head(1byte)	FrameID(1byte)	PEC(1byte)	Control Byte(1byte)			Data(Data Lengthbyte)
				Data Length:bit7:4	Frame Type:bit3:1	Checksum Type: bit0	
ACK	**0xA5**	**FID**		0	0	Checksum	
	Head(1byte)	FrameID(1byte)	PEC(1byte)	Control Byte(1byte)			Data(Data Lengthbyte)
				Data Length:bit7:4	Frame Type:bit3:1	Checksum Type: bit0	
RSP	**0xA5**	**FID**		Len	0	Checksum	Data[0:Len-1]